Dear HLDM 2026 Workshop Chairs,

We are pleased to submit our manuscript entitled *"From Multimodal Observation to Interpretable Suggestions: Counterfactual Time-Expanded Relational Modeling of Surgical Teams"* for consideration for presentation at the ECMLPKDD 2026 Workshop on Hybrid Human-Machine Learning and Decision Making (HLDM'26).

Our work directly aligns with the workshop's goal of advancing hybrid human-AI systems that go beyond prediction and actively support human decision-making in complex, high-stakes environments. In particular, we address the challenge of modeling and improving collaborative behavior in surgical teams, in which effective human-machine integration can have a direct impact on safety and performance.

In this paper, we propose a tempo-relational framework for multimodal surgical team modeling based on Time-Expanded Relational Neural Networks (TE-ReNN). Our approach jointly captures temporal evolution and interpersonal interactions within a unified graph structure, enabling robust learning even in low-data. Beyond predictive modeling, we introduce a counterfactual reasoning mechanism that generates interpretable and actionable suggestions for improving teamwork. These suggestions, grounded in multimodal data, identify minimal, structured changes in both interaction patterns and behavioral features that are associated with improved team outcomes.

The main contributions of our work are:

- A Time-Expanded Relational Neural Network (TE-ReNN) that integrates multimodal, temporal, and relational information for surgical team modeling in low-data regimes.
- A novel counterfactual reasoning mechanism that translates model predictions into interpretable and actionable recommendations at both behavioral and interaction levels.
- An empirical evaluation on real surgical training data demonstrating improved predictive performance and the ability to generate meaningful, human-interpretable interventions for team improvement.

We believe this work is highly relevant to HLDM'26 as it contributes to the design of hybrid human-machine decision-making systems that not only predict outcomes but also provide structured, actionable feedback to support human collaboration. In particular, our counterfactual mechanism is explicitly designed to bridge machine learning predictions and human decision processes, making it suitable for integration in human-in-the-loop settings such as surgical training and assessment.

We confirm that this submission is currently under review at another venue, for this reason, we are going to apply for the non-archival submission.

We thank you for your consideration and look forward to the opportunity to contribute to the workshop.

Sincerely,
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